

AI & Software Engineering Career Roadmap

3rd-Year B.Tech IT Student · Target: SWE & AI/ML Internships · Timeline: 12 Months

■ CURRENT STATUS		■ TARGET GOAL	
■ Student Profile	3rd-Year B.Tech IT	■ Primary Goal	SWE + AI/ML Internship
■ Languages	C++, Java, Python, SQL	■ Companies	Product, Startup, Big Tech
■ CS Fundamentals	DSA, OOP, DBMS basics	■ Timeline	12 months (by mid-2027)
■ Current Tools	Canva, basic AI tools	■ Package Target	■8–20 LPA (intern/PPO)
■ Time Available	~3–4 hrs/day	■ Placement Prep	Start Month 7 onwards

■ SKILL GAP ANALYSIS

Skill Area	Current Level	Target Level	Difficulty	Duration
Machine Learning / AI	Beginner	Intermediate	■ HIGH	3 months
System Design	None	Basic Concepts	■ MED	2 months
Web Dev (Backend API)	Minimal	REST APIs, FastAPI	■ HIGH	2 months
GitHub / DevOps Basics	Basic	CI/CD, Docker basics	■ MED	1 month
DSA (Advanced)	Intermediate	LeetCode Top 150	■ HIGH	Ongoing
LLMs & Prompt Engg.	None	Applied LLM Dev	■ MOD	1.5 months

■ LEARNING PLAN

Phase	Focus Areas	Project	Stack / Impact
M1–2 Foundations	Git, Python libs (NumPy, Pandas), FastAPI basics, LeetCode Easy-Med	■ AI Resume Screener	Python, NLP, FastAPI · HR-tech
M3–4 ML Core	Scikit-learn, regression/classification, Kaggle, HuggingFace intro	■ Stock Predictor ML	Scikit-learn, Streamlit · Kaggle
M5–6 AI / LLM	LangChain, RAG pipelines, Prompt Engineering, APIs, Project sprint	■ RAG Chatbot	LangChain, GPT, FAISS · LLM
M7–9 Placement	System Design, LeetCode Hard, mock interviews, resume polish	■ Placement Portal	Django, SQL, REST · Full-stack
M10–12 Apply	Internship applications, referrals, open-source contributions	■ DSA Visualiser	React, Python · Open-source

■ MONTHLY MILESTONES

GitHub profile live Python libs mastered 50 LeetCode solved First mini-project	2 Kaggle notebooks ML models deployed LinkedIn optimised Contribute to OSS	RAG chatbot live Blog post published Hackathon attended 100 LC solved	System Design basics Mock interviews x10 Portfolio polished 200 LC solved	10+ apps sent Referrals activated Offer in hand PPO target hit

■ NETWORKING & ACTION PLAN

Platform	Action	Immediate Next Action
LinkedIn	Post projects weekly, connect with SWEs & recruiters	■ Create GitHub profile & push first repo today
GitHub	Daily green squares, star & fork ML repos	■ Set up LinkedIn, add skills & profile picture
Twitter/X	Follow AI researchers, share learnings publicly	■ Solve 1 LeetCode Easy problem daily (streak!)
Discord	Join AI/ML communities, ask & answer questions	■ Enrol: Andrew Ng ML Specialisation (Coursera)
Referrals	Reach alumni on LinkedIn for internal referrals	■ Start 60-Day AI Challenge tracker
		■ Join 2 Discord AI/dev communities this week